SQL PORTFOLIO PROJECT

PPV Analysis
Production Performance Variance

Oracle Fusion | Manufacturing | Variance Analysis | CTEs | Window Functions

Project Summary

This query calculates Production Performance Variance (PPV) -- the difference between planned and actual cases produced per work order -- across multiple plants. It layers in rework counts (dented/scrapped cases), container size attributes, and package yield metrics, giving finance and operations a single source of truth for production efficiency reporting.

Thomas Foster

Data Analyst | Oracle Fusion ERP | SQL

github.com/thomasfoster | thomasfoster79@yahoo.com

1 PROJECT OVERVIEW

Platform:	Oracle Fusion ERP (Oracle SQL / PL-SQL)
Query Type:	Multi-CTE analytical query (8 CTEs + final SELECT)
Grain:	Work-order / transaction-date / item
Key Outputs:	Cases produced, planned cases, variance, package yield %
Parameters:	:PLANT_CODE, :P_START_DATE, :P_END_DATE, :item_number, :WORK_ORDER_NUMBER
Audience:	Finance, Operations, Supply Chain

2 BUSINESS PROBLEM

Production teams needed a reliable, drill-through report to answer: 'How much did we produce vs. what we planned -- and where did we lose cases to rework?' Without a single query joining work orders, transaction logs, planned quantities, and quality dispositions, analysts were reconciling four separate Excel exports -- a process that took hours and introduced reconciliation errors.

- Finance required weekly variance at the item/plant level for cost accruals.
- Operations needed overall package yield (good cases divided by total output) to spot quality trends.
- Planning used production variance to adjust future work-order quantities.
- The report had to handle multi-plant deployments with a single parameterized query.

3 TECHNICAL EXPLANATION

The query chains eight CTEs that progressively enrich the data before the final SELECT:

- ORG_DATA -- filters to the selected plant(s) from INV_ORG_PARAMETERS.
- ITEM_BASE -- restricts items to CASE/CS unit-of-measure for can-line relevance.
- CS_SET / CS_ASSIGN_ALL / CS_PICK -- resolve container-size categories using ROW_NUMBER() to pick the best match per item (plant-specific preferred over global).
- REWORK_COUNTS_ALL / REWORK_PICK -- aggregate dented and scrapped case counts from quality category assignments using DENSE_RANK KEEP.
- WO_BASE -- joins work orders to the MSS production plan and transaction log, applying all user parameters (date range, item, work order number).
- TXN_SUMMARY -- pivots OP_COMPLETION transaction types into cases_produced grouped by week and work order.
- PLANNED_QTY -- sums planned material quantities from work-order operation materials.
- Final SELECT -- joins all CTEs; computes variance, variance %, package yield, and timestamps in US/Central time.

Key Techniques: CTEs, ROW_NUMBER(), DENSE_RANK KEEP, conditional aggregation, timezone conversion, NVL

4 MASKED SQL -- KEY CTE EXCERPTS

```
-- ORG_DATA: plant filter
WITH ORG_DATA AS (
  SELECT DISTINCT op.organization_id, op.organization_code
  FROM   inv_org_parameters op
  WHERE  (:PLANT_CODE IS NULL OR op.organization_code = :PLANT_CODE)
),
-- CS_PICK: best container-size per item using ROW_NUMBER()
CS_PICK AS (
  SELECT * FROM (
    SELECT csa.*,
           ROW_NUMBER() OVER (
             PARTITION BY csa.inventory_item_id
             ORDER BY CASE WHEN csa.organization_id = :PLANT_CODE THEN 0
                           WHEN csa.organization_id IS NULL        THEN 1
                           ELSE 2 END, csa.can_size_code) rn
    FROM CS_ASSIGN_ALL csa
  ) WHERE rn = 1
),
-- TXN_SUMMARY: pivot OP_COMPLETION to cases_produced
TXN_SUMMARY AS (
  SELECT wb.transaction_week, wb.work_order_number, od.organization_code,
         wb.item_number, wb.item_description,
         SUM(CASE WHEN tx.transaction_type_code = 'OP_COMPLETION'
                  THEN tx.transaction_quantity ELSE 0 END) AS cases_produced
  FROM WO_BASE wb
  JOIN cst_wo_operation_txns tx
    ON wb.work_order_id = tx.work_order_id
   AND TRUNC(tx.transaction_date,'IW') = wb.transaction_week
  JOIN ORG_DATA od ON wb.organization_id = od.organization_id
  GROUP BY wb.transaction_week, wb.work_order_number,
           od.organization_code, wb.item_number, wb.item_description
),
-- Final SELECT: variance and package yield
SELECT
  t.organization_code, t.item_number, t.item_description,
  t.cases_produced,
  NVL(p.planned_cases, 0)                                          AS planned_cases,
  (t.cases_produced - NVL(p.planned_cases,0))                      AS total_variance,
  ROUND((t.cases_produced - p.planned_cases)
        / NULLIF(p.planned_cases,0), 4)                            AS variance_pct,
  ROUND(t.cases_produced
        / NULLIF(t.cases_produced + NVL(rw.cases_scrapped,0)
                                  + NVL(rw.cases_dented,0), 0), 4) AS pkg_yield
FROM TXN_SUMMARY t
LEFT JOIN PLANNED_QTY  p  ON p.work_order_id     = t.work_order_id
LEFT JOIN CS_PICK      cs ON cs.inventory_item_id = t.inventory_item_id
LEFT JOIN REWORK_PICK  rw ON rw.inventory_item_id = t.inventory_item_id
```

5 EXAMPLE OUTPUT

Plant	Item #	Description	Produced	Planned	Variance	Yield %

[PLANT_A]	ITEM-001	Canned Tomatoes 14oz	4,820	5,000	-180	97.2%
[PLANT_A]	ITEM-002	Kidney Beans 15oz	3,102	3,000	+102	98.8%
[PLANT_B]	ITEM-003	Corn Whole Kernel 15oz	2,950	3,100	-150	96.1%
[PLANT_B]	ITEM-004	Green Beans Cut 14.5oz	1,880	1,900	-20	99.0%

** Values illustrative. Organization codes and item numbers masked.*